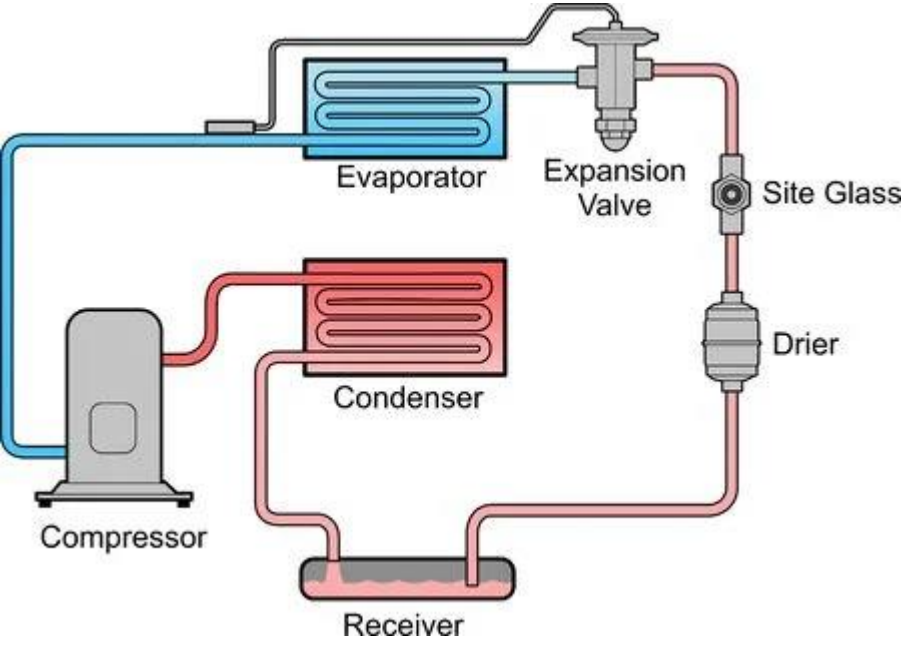
Mechanical Engineering Department, CoE, KNUST

ME 465: Refrigeration and Air-Conditioning

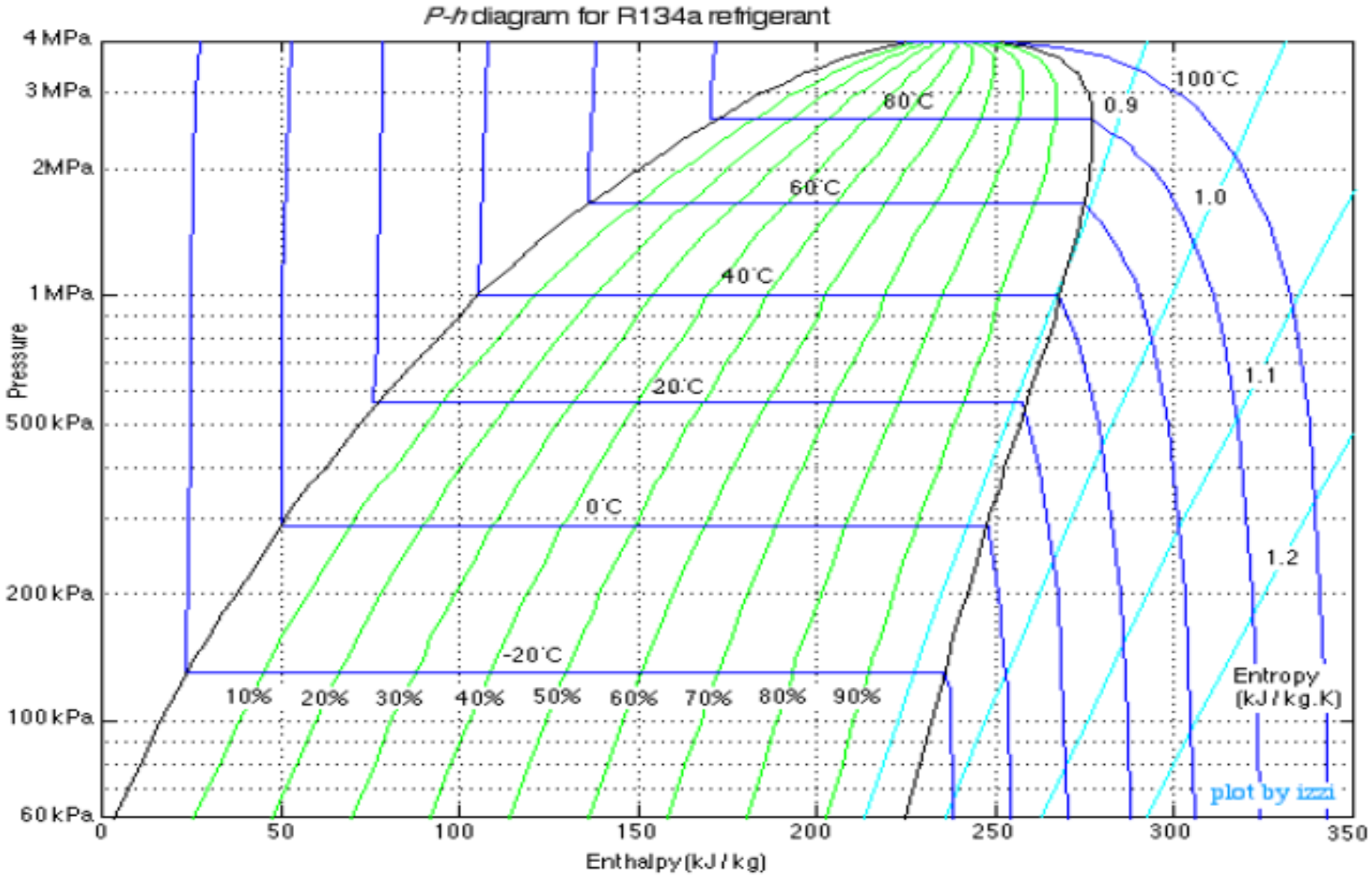
LECTURE 2



Course Lecturer:
Dr. C. Sekyere

Examples on the Mechanical Vapour Compression Cycle

The P-h diagram for R134a



Examples on the Mechanical Vapour Compression Cycle

EXAMPLE 1

A refrigeration plant operating with a piston compressor and refrigerant R134a shall be built. The required cooling capacity is 25 kW at evaporating temperature of minus 20 ^{o}C. Suction superheat at exit of evaporator shall be 10 ^{o}C. Condensing temperature shall be +40 ^{o}C and the condensate is sub-cooled in the condenser by 10 ^{o}C. Assuming that the isentropic and volumetric efficiencies of compressor are 65% and 60% respectively and that the heat loss from the compressor corresponds to 15% of the input shaft power, calculate the following:

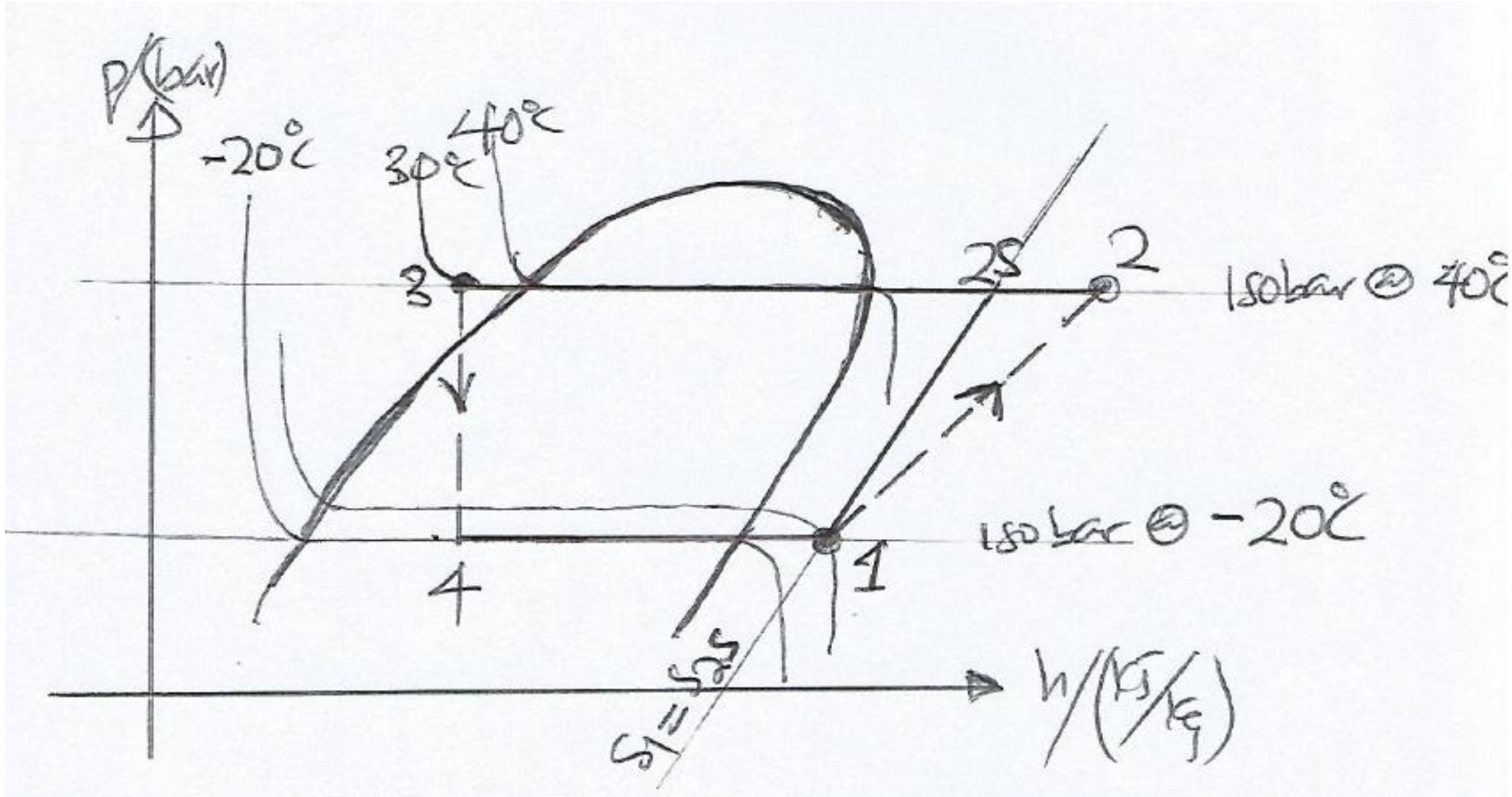
(i)Required swept volume

(ii)Input power of compressor

(iii)Condenser heat load

(iv)COP of plant

SOLUTION



From the p – h chart of R 134a, the following parameters are read:
State point 1, intersection of isobar for -20 °C sat. and T = -10 °C isotherm: h1 = 395 kJ/kg
State point 2s, intersection of constant "s" drawn from 1 and isobar for +40 °C: h2s = 440 kJ/kg
State 3, intersection of isobar for +40 °C sat. and T= +30 °C isotherm: h3 = h4 = 245 kJ/kg

State point 1: $v_1 = 0.15\ m^3/kg$

Circulated mass: $\dot{m} = \frac{\dot{Q}}{h_1 - h_4} = \frac{25}{440-245} = 0.1282\ kg/s$

Volume induced: $\dot{V}_{in} = \dot{m}_{ref} v_1 = 0.1282 \times 0.15 = 0.00192\ m^3/s$

(i) Required swept volume: $\dot{V}_s = \frac{\dot{V}_{in}}{\eta_{vol}} = \frac{0.0192}{0.60} = 0.0295\ m^3/s$

(ii) Shaft power: $w_r = \frac{w_{is}}{\eta_{isen,comp}} = \frac{\dot{m}_{ref} \times (h_{2s} - h_1)}{\eta_{isen,comp}} = \frac{(440-395)}{0.65} = 9.6\ kW$

(iii)Condenser load: $Q_c = Q_e + P_r - Q_{loss} = 25 + (1 - 0.15) \times 9.6 = 33.2\ kW$

(iv) Coefficient of Performance: $COP_{ref} = \frac{Q_e}{W_r} = \frac{25}{9.6} = 2.6$

EXAMPLE 2

A cold store maintains a storage temperature of minus 23 ^{o}C in an ambient of 27 ^{o}C. The plant is fitted with a single 4-cylinder compressor having a bore of 150 mm, stroke 100 mm, speed 25 rev/sec, and volumetric efficiency of 80%. It operates with refrigerant 134a and has a Carnot efficiency of 50%. The discharge gas temperature is 80 ^{o}C, evaporating temperature minus 30 ^{o}C, condensing temperature 40 ^{o}C and suction gas superheat 10 ^{o}C. Determine the following:

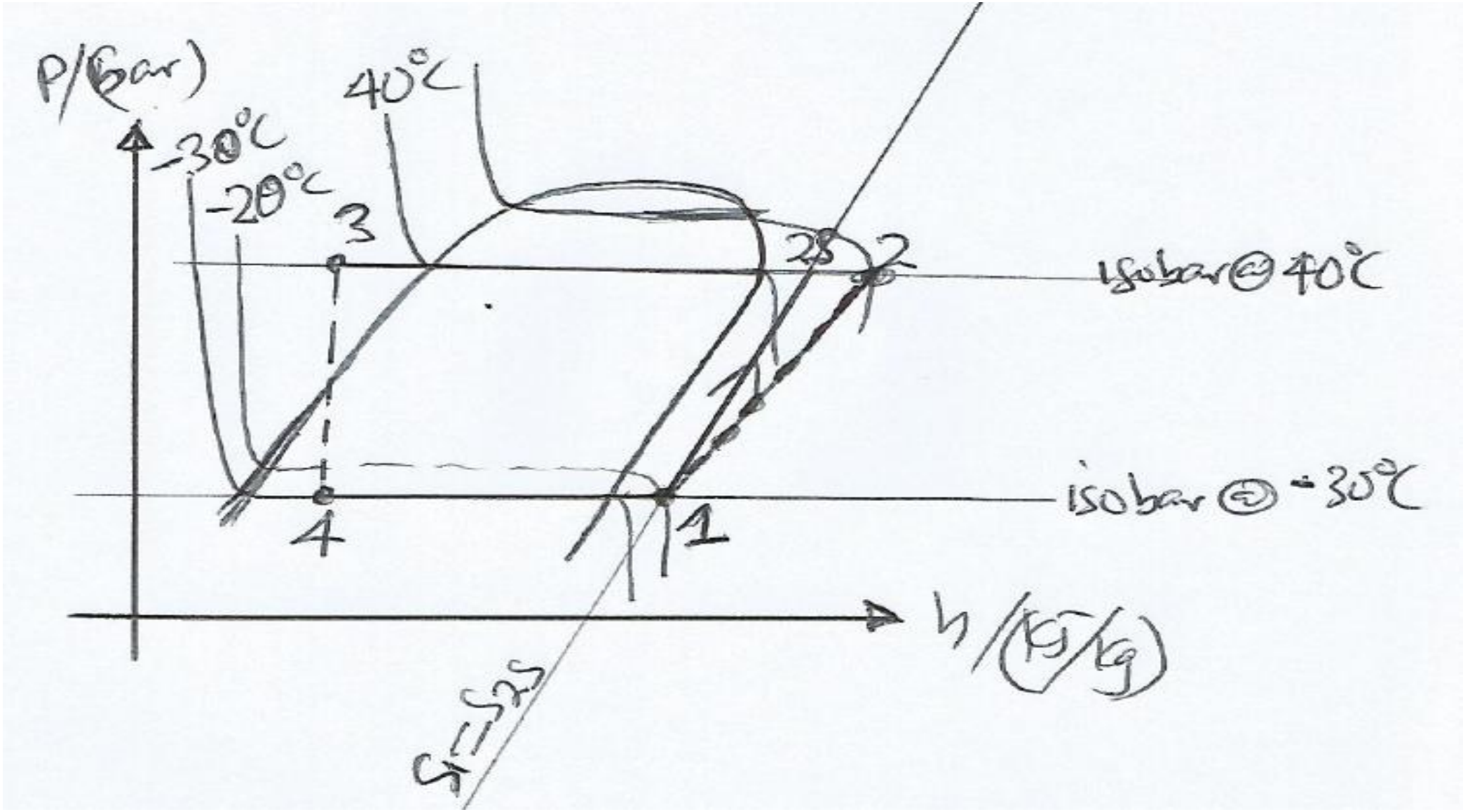
(i)Isentropic efficiency of compressor

(ii)Power consumption of compressor

(iii)Condensate sub-cooling

(iv)If the compressor is driven by an electric motor of 80% efficiency and electricity is produced from primary fuel with efficiency of 40%, what is the amount of primary fuel energy input, in kJ per hour, required to run the plant.

SOLUTION



SOLUTION

From the p – h chart of R 134a, the following parameters are read:
State point 1, intersection of isobar for -30 °C sat. and T = -20 °C isotherm: h1 = 390 kJ/kg
State point 2s, intersection of constant "s" drawn from 1 and isobar for +40 °C: h2s = 440 kJ/kg
State 2, intersection of +80 °C isotherm and +40 °C sat isobar: h2 = 452 kJ/kg

State point 1: $v_1 = 0.15\ m^3/kg$

(i)Isentropic efficiency, $\eta_{isen,comp} = \frac{(h_{2s} - h_1)}{(h_2 - h_1)} = \frac{(440-390)}{(452-390)} = 83.9\%$

(ii) Compressor displacement: $\dot{V}_s = \frac{\pi d^2}{4} LnN = \frac{\pi \times 0.15^2}{4} \times 0.1 \times 4 \times 25 = 0.1767\ m^3/s$

Induced volume: $\dot{V}_r = \eta_{vol} \times \dot{V}_s = 0.8 \times 0.1767 \times 4 \times 25 = 0.1414\ m^3/s$

Induced mass: $\dot{m} = \frac{\dot{V}_r}{v_1} = \frac{0.1414}{0.25} = 0.5657\ kg/s$

Compressor power: $P = \dot{m} \times (h_2 - h_1) = 0.5657 \times (452 - 440) = 35\ kW$

(iii) Condensate sub-cooling: $COP_{carnot} = \frac{273 + (-23)}{27 - (-23)} = 5$

SOLUTION

$$COP_{plant} = \eta_{Carnot} \times COP_{Carnot} = 0.50 \times 5 = 2.5$$

Now, $$COP_{plant} = \frac{h_1 - h_4}{h_2 - h_1} = \frac{390 - h_4}{452 - 390} \quad \rightarrow \quad h_4 = h_3 = 235\ kJ/kg$$

Condensate sub-cooling: $\Delta T = 40 - 25 = 15\ ^{o}C$

(iv) Primary fuel energy, PFE: PFE×0.40×0.80 = 35 kW
PFE = 103 kW OR 370,800 kJ/h

Thank you